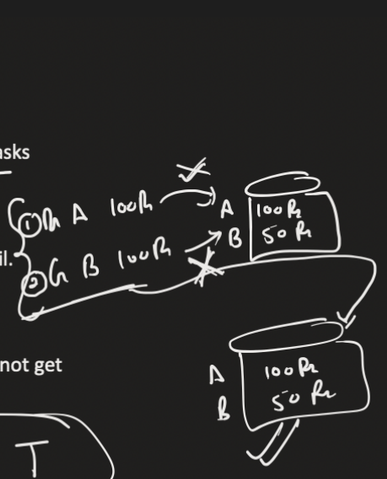
HLD: Distributed Transaction Handling (2PC, 3PC and SAGA)

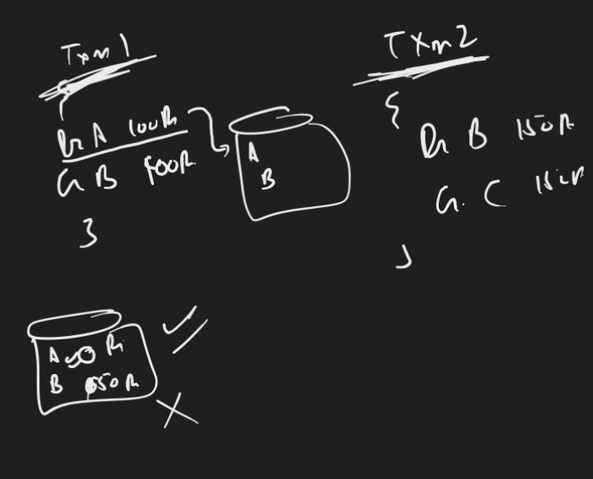
"Concept && Coding" YT Video Notes

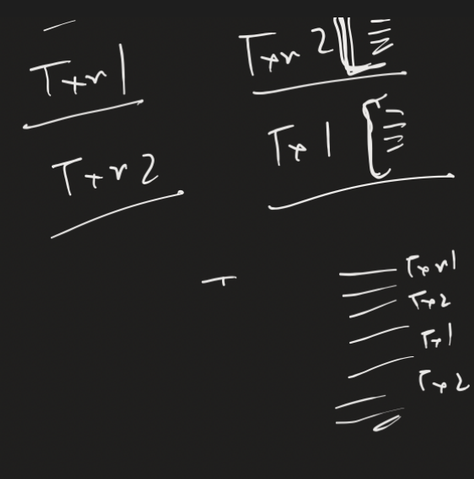
What is Transaction:

It refers to a set of operations which need to be performed or simply say group of tasks which need to be performed against the DB.

It has 4 properties: ACID
- Atomicity: All operations in a single transaction should be success or all should fail.
- Consistency: DB should be in consistent state before and after the transaction.
- Isolation: More than one transaction that is running concurrently, appears to be serialized.
- Durability: After transaction successfully completed, even if DB fails, data should not get lost.



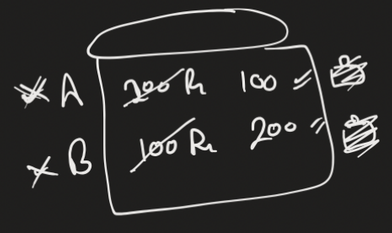





```
Txn
Begin_Transaction
    - Withdrawal Money from 'A': Rs 100
    - Deposit Money to 'B': Rs 100

If all success:
        Do Commit
else
        Do Rollback

Close_Transaction;
```

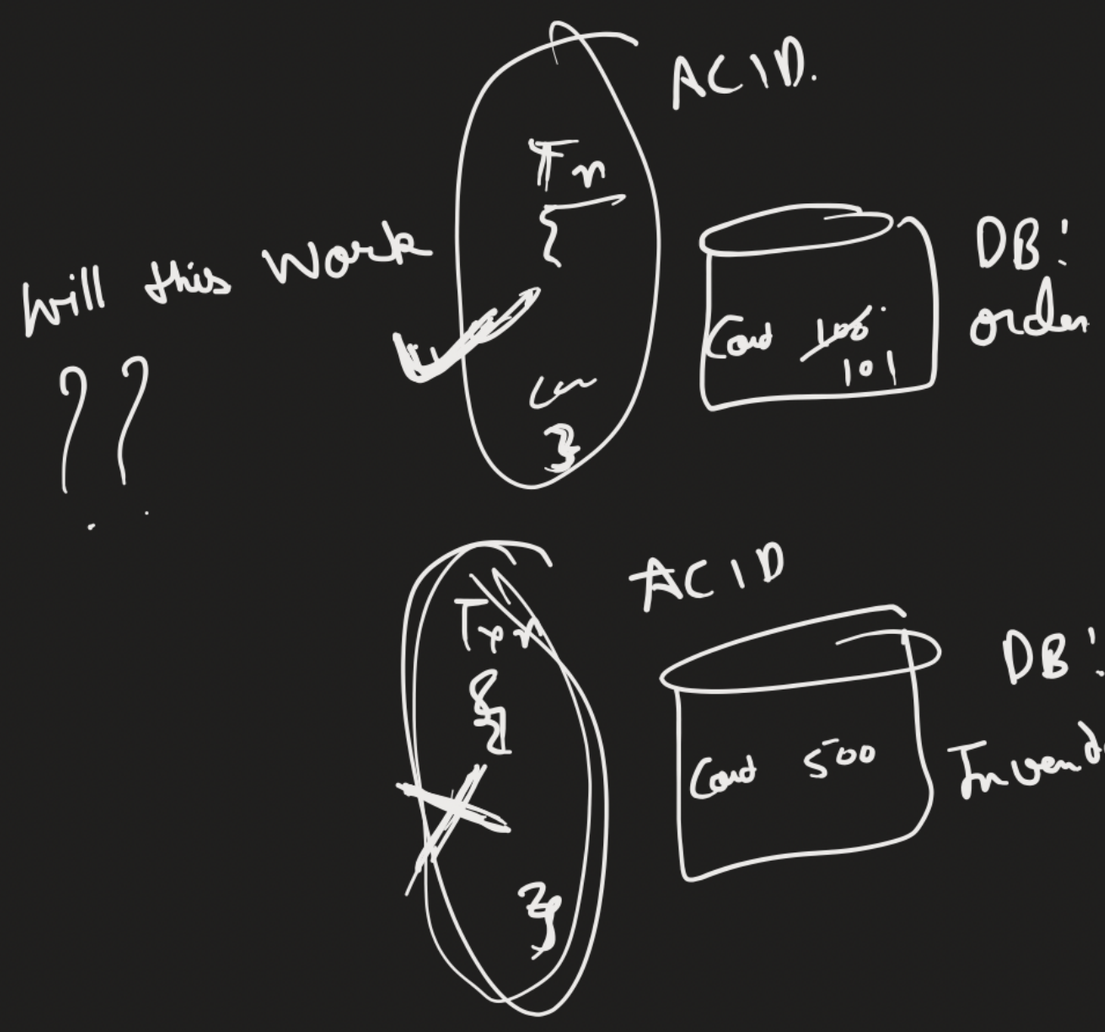


How to handle this in Distributed System, where operations involves multiple databases?
or
As Transaction is Local to a particular database? How we will satisfy the transaction Property in Distributed system?

```
Begin_Transaction:
    Update Order DB
    Update Inventory DB

If all success:
        Do Commit
else
        Do Rollback

Close_Transaction;
```



How to handle Transaction in Distributed System?

